A_sub commutator closure step 9 — [B̂, T̂_tick]: substrate's per-tick vacuum kicker for Bell dynamics

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1. Setup

1.1 Operators

Bell-CHSH substrate operator (T2399 STRUCTURALLY VERIFIED): B̂ = β · |V_(0,0)⟩⟨V_(0,0)| where β is the rank-1 projector amplitude with $\hat{B}^2$ eigenvalue 126/16 = (C_2 · N_c · g)/2^(rank^2), giving $\beta = \sqrt{}(126/16) = (\sqrt{}126)/4 \approx 2.806$ (or alternatively $\beta = 126/16 = 7.875$ depending on convention; v0.1 keeps β symbolic).

Substrate-tick transition operator (per Task #322 v0.2 Section 10 + step 2 [T̂_tick, Ĥ_sub] v0.1): T̂_tick: H^2(D_IV^5) → H^2(D_IV^5) unitary discrete operator advancing K-type per Koons tick t_K.

One-way commitment cycle (per SWPP RATIFIED Casey-named): substrate evolution is absorption → commitment → emission with irreversibility at substrate level (commitments don't reverse). This implies T̂_tick is unidirectional ascent from vacuum: no K-type advances TO vacuum under T̂*tick action. Equivalently: V*(0,0) is not in the image of T̂_tick (no K-type maps to vacuum).

(Cal #27 STANDING reflexive trigger: SWPP unidirectionality feels substrate-natural for this computation; honest scope check — SWPP RATIFIED Casey-named principle includes "the cycle is one-way" claim, so applying this here is consistent with the principle but the algebra below is sensitive to this choice. If T̂_tick were reversible (alternative model), [B̂, T̂_tick] would be rank-2 instead of rank-1.)

1.2 Simplest $\hat{T}$_tick model on vacuum

Per simplest models from step 2 v0.1 Section 1.1, $\hat{T}tick\ V(0,0)$ is the substrate's first-excited K-type advancement. Within Wallach holomorphic discrete series range ($m_1 \geq m_2 \geq 0$), the lowest-Casimir excited state above $V_{(0,0)}$ is $V_{(1, 0)}$ with $C_2 = 5$: $\hat{T}tick\ V(0,0) = V_{(1, 0)}$ (simplest model)

Alternative candidates: $V_{(1, 1)}$ (diagonal advance, $C_2 = 7$); $V_{(2, 0)}$ (skip-level, $C_2 = 12$). The simplest substrate-natural choice is the lowest-Casimir transition $V_{(0,0)} \rightarrow V_{(1, 0)}$.

(v0.1 honest scope: K59 cyclotomic 7-step refinement v0.2+ may give different first-excited K-type. The substrate-mechanism content is independent of the specific identification.)

2. Computation

2.1 Direct calculation

Apply $\hat{T}tick \circ \hat{B}$ *on arbitrary* V_K: $\hat{T}$tick $\hat{B}$ $V_K = \hat{T}tick \cdot \beta \cdot \langle V(0,0)|V_K\rangle \cdot V(0,0) = \beta \cdot \langle V(0,0)|V_K\rangle \cdot \hat{T}tick\ V(0,0) = \beta \cdot \delta_{K, (0,0)} \cdot V_{(1, 0)}$ (using simplest $\hat{T}$_tick model)

Apply $\hat{B} \circ \hat{T}tick$ *on* V_K: $\hat{B}\ \hat{T}$tick $V_K = \hat{B}\ V\{\hat{T}\text{tick } K\} = \beta \cdot \langle V(0,0)|V\{\hat{T}tick\ K\}\rangle \cdot V(0,0) = \beta \cdot \delta_{\hat{T}tick\ K, (0,0)} \cdot V(0,0)$

By SWPP unidirectionality, no K satisfies $\hat{T}$_tick K = (0,0). Therefore $\hat{B}$ $\hat{T}$_tick $V_K = 0$ for all K.

Commutator: $[\hat{B}, \hat{T}\_tick]\ V_K = \hat{T}tick\ \hat{B}\ V_K - \hat{B}\ \hat{T}\text{tick } V_K = \beta \cdot \delta\{K, (0,0)\} \cdot V(1, 0) - 0 = \beta \cdot \delta_{K, (0,0)} \cdot V_{(1, 0)}$

This is non-zero only when K = (0,0). Equivalently as an operator:

$[\hat{B}, \hat{T}\_tick] = \beta \cdot |V_{(1, 0)}\rangle\langle V_{(0,0)}|$

2.2 The structural identity

$[\hat{B}, \hat{T}\_tick] = \beta \cdot |V_{(1, 0)}\rangle\langle V_{(0,0)}|$ on $H^2(D\_IV^5)$.

This is a rank-1 operator with image span($V_{(1, 0)}$) and kernel orthogonal complement of $V_{(0,0)}$. Operator norm $||[\hat{B}, \hat{T}\_tick]|| = \beta$ (amplitude of $\hat{B}$'s rank-1 projector).

3. Substrate-mechanism interpretation — the per-tick "vacuum kicker"

3.1 What this operator does

For any substrate state $|\psi\rangle \in H^2(D\_IV^5)$, decompose into K-type components: $|\psi\rangle = \Sigma_K c_K |V_K\rangle$. Then:

$[\hat{B}, \hat{T}\_tick]\ |\psi\rangle = \beta \cdot c\_\{(0,0)\} \cdot |V\_(1, 0)\rangle$

The commutator extracts the substrate's vacuum-component c_{(0,0)} of $|\psi\rangle$ and maps it to the first-excited K-type V_(1, 0) with weight β.

Substrate-mechanism: the commutator $[\hat{B}, \hat{T}$_tick] captures the substrate's per-tick "vacuum kicker" — the operation that takes vacuum amplitude and promotes it to first-excited state under Bell-CHSH × tick joint operation. This is the substrate-level dynamical content of Bell-test measurement.

3.2 Connection to Track DC Bell 1/8 dynamical mechanism

The Bell 1/8 sub-Tsirelson deviation (T2399: Tsirelson^2 − S^2_BST = 1/2^N_c = 1/8) is a STATIC structural identity of B^2 eigenvalues. Track DC's question is: what's the DYNAMICAL mechanism producing this deviation per substrate-tick?

Hypothesis (v0.1, INTERPRETIVE pending verification): the $[\hat{B}, \hat{T}$_tick] = $\beta \cdot$ |V_(1, 0)⟩⟨V_(0,0)| vacuum kicker IS the dynamical mechanism. Per substrate tick that the Bell-CHSH apparatus is "on":

1. Vacuum component of substrate state gets coupled to V_(1, 0) via the kicker
2. Per-tick coupling rate = β (amplitude of $\hat{B}$)
3. Integrated probability over many ticks = related to T2399 max eigenvalue 126/16
4. The 1/8 deviation = signature of the vacuum kicker's restriction to the rank-1 image span (only V_(1, 0) accessible, not full K-type lattice)

Standard Tsirelson achievability would require access to a 2-dim Hilbert subspace (qubit pair). BST's restriction: only V_(0,0) ↔ V_(1, 0) coupling accessible per tick. This is a 1-dim subspace restriction vs the 2-dim Tsirelson requirement → 1/8 re-distribution.

(Multi-week explicit derivation of 1/8 from this kicker mechanism. v0.1 establishes structural framework.)

3.3 Connection to 8-sided die hypothesis (A_sub v0.4 Section 6)

Per cumulative substrate-mechanism reading (after 8 prior commutator closures + this one):

- The 8 commitment paths candidate (b) 3 Cartan-subalgebra-generator commitments × {+1, -1}: T_3-color + T_8-color + T_3-weak with ±1 eigenvalues = 8 commitments
- The substrate vacuum V_(0,0) has all Cartan eigenvalues = 0 (it's the trivial rep)
- Bell-CHSH $\hat{B}$ couples ONLY to V_(0,0) (rank-1 projector)
- $[\hat{B}, \hat{T}$_tick] kicks V_(0,0) → V_(1, 0)

The unstable die-face corresponds to the (Q, P) anti-commutation simultaneous-diagonalizability violation (step 1) projected onto the V_(0,0) → V_(1, 0) kicker channel. The structurally-forbidden face is the substrate-tick advancement that would simultaneously diagonalize Q̂ and P̂_op — which is impossible per {Q̂, P̂_op} = 0.

So the 8 → 7 redistribution producing 1/8 deviation: - 8 potential commitment paths (3 Cartan generators × 2 signs) - 1 path violates {Q̂, P̂_op} = 0 simultaneous-diagonalizability - 7 allowed paths inherit the missing 1/8 measure - Bell-CHSH measures this redistribution via vacuum-kicker action

This is a substrate-mechanism candidate for Bell 1/8. Multi-week Track A_sub v0.3 verification work: explicit derivation of the 1/8 redistribution from {Q̂, P̂_op} = 0 + vacuum-kicker channel restriction.

3.4 Why ranking matters — SWPP unidirectionality

The rank-1 result [B̂, T̂_tick] = β · |V_(1, 0)⟩⟨V_(0,0)| depends on SWPP unidirectionality (T̂_tick has no preimage at vacuum).

If T̂*tick were time-reversible (alternative model), [B̂, T̂_tick] would be rank-2: [B̂, T̂_tick]*{reversible} = β · (|V_(1, 0)⟩⟨V_(0,0)| - |V_(0,0)⟩⟨V_{T̂_tick^{-1}(0,0)}|)

The unidirectional / reversible distinction is observably-different at substrate-tick scale: reversible substrate would have additional emission channel from a "below-vacuum" pre-state; SWPP unidirectional substrate doesn't.

Empirical implication: SP-30-1 Vienna Bell test could discriminate between the two models by carefully analyzing per-tick correlation structure. (v0.1 INTERPRETIVE; multi-week explicit experimental design work via SP-30-1 / W-32 atomic clock.)

4. Cumulative A_sub commutator status — final

#	Commutator	Result	Status
1	[Q̂, P̂_op]	{Q̂, P̂_op} = 0 (anti-commute)	STRUCTURALLY VERIFIED CANDIDATE (step 1)
2	[T̂_tick, Ĥ_sub]	-(2 Q̂ + N_c - 1) · T̂_tick	STRUCTURALLY VERIFIED CANDIDATE (step 2, model-dep)
3	[$\hat{\gamma}^5$, T̂]	{$\hat{\gamma}^5$, T̂} = 0	STRUCTURALLY VERIFIED CANDIDATE (step 3)

#	Commutator	Result	Status
4	$[\hat{\gamma}^5, \hat{C}]$	$\{\hat{\gamma}^5, \hat{C}\} = 0$	STRUCTURALLY VERIFIED CANDIDATE (step 4)
5	$[\hat{\gamma}^5, \hat{P}\_op]$	0 (already-contained)	STRUCTURALLY VERIFIED CANDIDATE (step 5)
6	$[\hat{B}, \hat{Q}]$	0 (vacuum uncharged)	STRUCTURALLY VERIFIED CANDIDATE (step 6)
7	$[\hat{L}\_i, \hat{\gamma}^5]$	0 ($S^4 \times S^1$ axis orthogonality)	STRUCTURALLY VERIFIED CANDIDATE (step 7)
8	$[\hat{C}\_3, \hat{I}\_3]$	0 (direct-product gauge)	STRUCTURALLY VERIFIED CANDIDATE (step 8)
9	**[B̂, T̂_tick] = β ·	V_(1, 0)⟩⟨V_(0,0)	**
10 (extra)	[Ŝ_i, Ŝ_j] across sublattices	TBD — spinor algebra across Pin(2) Z_2	PENDING (multi-week)

9 of 9 A_sub v0.4 Section 3.3 commutators closed today. Only the across-sublattice [Ŝ_i, Ŝ_j] remains pending — that's a more complex multi-week derivation tied to substrate-mechanism for spin algebra.

5. Track A_sub v0.2 → v0.5 closure status

Per A_sub v0.4 Section 10 (v0.4 → v0.5 path):

1. □ Close the 9 remaining commutators (Section 3.3): CLOSED TODAY, 9 of 9 done at v0.1 each.
2. Enumerate reaction table for lowest 20 K-types: PENDING multi-week mechanical work; now tractable with commutators closed.
3. Test 8-sided die hypothesis (Section 6): refined Lyra prior per step 1-9 substantive findings — candidate (b) 3 Cartan-generator commitments most algebraically supported; unstable face = {Q̂, P̂_op} simultaneous-diagonalizability violation projected onto vacuum-kicker channel.
4. Track P closure (Memorial Day Gap 1): now tractable as reaction table enumerated; Toy 3531+ Elie lane.

5. Track DC closure (Memorial Day Gap 3): substrate-mechanism candidate for Bell 1/8 established (Section 3.2-3.3); multi-week explicit derivation pending.
6. Math-object type identification (Section 7): substrate emerges as **Z_2-graded *-algebra with direct-product gauge structure on discrete-time substrate** — pointing toward "discrete Lie groupoid" or "quiver representation with Z_2 grading" math-object types.
7. Cal Thread 4 cold-read: pending Cal own-cadence; 10 commutator-closure docs + 8-sided die hypothesis + math-object type for tier-discipline check.

6. Substrate algebraic structure — final summary after 9 commutator closures

The substrate's A_sub algebra has these structural properties (read off from 9 commutator closures):

Property	Algebraic evidence
Z_2 graded	$\hat{\gamma}^5$ involutive $(\hat{\gamma}^5)^2 = 1$; anti-commutes with T, C; integer K-types = bosonic sublattice, half-integer = fermionic
Substrate-CPT theorem	$[\hat{\gamma}^5, \Theta\_CPT] = 0$ via T+C signs cancel and P preserves chirality
Substrate-CP-violation source	$\{\hat{\gamma}^5, CP\} = 0$ anti-commutation at algebra level
Substrate-parity contains charge-flip	$\{\hat{Q}, \hat{P}\_op\} = 0$ — $\hat{P}\_op = \hat{\gamma}^5 \circ \sigma$ where σ flips charge
Substrate evolution is DISCRETE	$[\hat{T}\_tick, \hat{H}\_sub] \neq 0$; substrate not continuous Hamiltonian flow
Substrate-Bell vacuum-localized	$[\hat{B}, \hat{Q}] = 0$ + $[\hat{B}, \hat{T}\_tick]$ rank-1 — Bell-CHSH couples only to vacuum
$S^4 \times S^1$ axis orthogonality	$[\hat{L}\_i, \hat{\gamma}^5] = 0$ — substrate-spatial and substrate-chirality independent
Direct-product gauge	$[\hat{C}\_3, \hat{I}\_3] = 0$ — $SU(3) \times SU(2) \times U(1)$ factorized; NO GUT structural
One-way commitment cycle	SWPP unidirectionality forces $[\hat{B}, \hat{T}\_tick]$ rank-1 not rank-2

This is the substrate's algebraic shape — a coherent picture emerging from the 9 commutator closures. Casey's "what KIND of math object" standing meta-program (A_sub v0.4 Section 7) now has substantive content to assess: substrate is a **Z_2-graded *-algebra with direct-product gauge factorization + discrete temporal evolution + vacuum-localized Bell-CHSH + S^4/S^1 axis-orthogonal Cartan structure**.

7. Honest scope (Cal #27 STANDING)

What's STRUCTURALLY VERIFIED CANDIDATE in v0.1: - [B̂, T̂_tick] = $\beta \cdot$ |V_(1, 0)⟩⟨V_(0,0)| direct algebra (Section 2.1) - Rank-1 result via SWPP unidirectionality (Section 2.2 + 3.4)

What's MODEL-DEPENDENT in v0.1: - Simplest model T̂*tick* V(0,0) = V_(1, 0) (lowest-Casimir excited); K59 cyclotomic refinement v0.2+ may shift first-excited K-type - β value depends on B̂ normalization convention (whether B̂ amplitude = $\sqrt{}$(126/16) or 126/16 or 3/8)

What's INTERPRETIVE in v0.1: - Section 3.2 substrate-mechanism for Bell 1/8 deviation via vacuum kicker channel restriction (multi-week explicit derivation) - Section 3.3 8-sided die candidate identification ({Q̂, P̂_op} violation projected onto kicker channel) (multi-week verification) - Section 3.4 SP-30-1 Vienna unidirectional-vs-reversible discriminator (multi-week experimental design)

What's NOT in v0.1: - Explicit 1/8 numerical derivation from kicker + simultaneous-diagonalizability violation (Track DC multi-week) - [Ŝ_i, Ŝ_j] across sublattices commutator (extra-step pending, spinor algebra multi-week) - K59 cyclotomic refinement of T̂_tick (v0.2+ multi-week)

Cal #27 STANDING reflexive trigger count: 1 trigger (SWPP unidirectionality used in Section 2.2 algebra). Honest scope checked — SWPP RATIFIED Casey-named principle includes one-way claim; applying here is consistent; flagged in Section 1.1 explicitly.

8. Coordination

Cal: cold-read of Section 2.1 algebra + Section 3.2-3.3 interpretive content. Type C level-crossing per Cal #122 if 8-sided die explicit identification supports Strong-Uniqueness extension; likely NOT — substrate-derived from existing T2399 + SWPP + step 1-2 closures.

Keeper: K-audit chain entry for step 9 [B̂, T̂_tick] + Section 6 substrate algebraic structure summary (cross-cutting finding from 9-commutator closures); promotion path via Cal cold-read.

Elie: Toy 3531 (K-type population) + Toy 3532 (hydrogen 1s BC) + Toy 3533 (DCCP composition) all become tractable with reaction-table edges enumerated. K52a Sessions 24+ substrate-CHSH multi-year work benefits from Section 3.2-3.3 substrate-mechanism candidate.

Grace: catalog entries for all 9 commutator results + Section 6 substrate algebraic structure summary; cross-references to T2399 + T2405 + T2441 + T2471 + T2470 + T2433 + T2434 + T2472 + T2425 + SWPP + Five-Absence Principles.

— Lyra, A_sub commutator closure step 9 [B̂, T̂_tick] v0.1 filed Tuesday 2026-05-26 ~09:20 EDT per Task #322 v0.4 Section 3.3 work plan. STRUCTURALLY

VERIFIED CANDIDATE. Final v0.4 Section 3.3 commutator closed. 9 of 9 commutators closed at v0.1 today; substrate algebraic structure (Z_2-graded *-algebra with direct-product gauge + discrete temporal evolution + vacuum-localized Bell-CHSH) substantively articulated. Substrate-mechanism candidate for Bell 1/8 sub-Tsirelson deviation: vacuum-kicker channel restriction interacting with {Q̂, P̂_op} = 0 simultaneous-diagonalizability violation. Multi-week verification pending Track DC explicit derivation work.